

Machine Learning Course 2024 Spring: Experiment 2

October 14, 2025

1 Overview

In this experiment, you will implement linear/non-linear Support Vector Machine (SVM) models on given data set, and deepen the knowledge and understanding of the hyperparameters and kernel functions involved through visualization.

2 Dataset Description

The datasets are ready for you and are stored in the file '*T1_data.csv*' and '*T2_data.csv*'. Each row in the files is a sample, with the last column being the label and the other columns being the features. You also need to load a real world dataset 'Wisconsin Breast Cancer Dataset' with the `sklearn` module.

3 Task Description

3.1 Task 1

Given the *T1* dataset, implement the **linear SVM** model. Visualize the decision boundary of the model. Tune the value of hyperparameter C (regularization parameter), and observe the change of decision boundary.

3.2 Task 2

- Given the *T2* dataset, implement the **non-linear SVM** model with **Gaussian kernel**. Visualize the decision boundary of the model. Tune the value of hyperparameter C and

γ , and observe the change of decision boundary.

- Tune the hyperparameter C and γ of the SVM model using a grid search. Use 21 values for C and γ in $[2^{-10}, 2^{-9}, \dots, 2^9, 2^{10}]$ on a log scale. Report accuracy under 5-fold cross validation and the optimal hyperparameter values.

3.3 Task 3

- Given the real world dataset split the dataset into a training set (80%) and a test set (20%). Standardize the features (e.g., using `StandardScaler`). Explain in your report why feature scaling is crucial for SVMs. Implement the **non-linear SVM** model with **Gaussian kernel**.
- Tune the hyperparameter C and γ of the SVM model using a grid search. Use the best model found to make predictions on the test set. Compute and report the following evaluation metrics: Accuracy, Precision, Recall, and F1-Score.

3.4 Task 4 (Optional)

Try to implement SVM without using existing machine learning libraries (`sklearn`, `LIBSVM`, `LIBLINEAR`, etc.).

4 Requirements

- Implement learning algorithms with **Python**. You can use integrated libraries such as `sklearn`, `LIBSVM`, `LIBLINEAR`, etc.
- Record the procedure of experiments in **Jupyter Notebook**. It is recommended to use **Markdown** in Notebook to properly explain the procedure.
- Your submission includes:
 - source code: `python notebook (.ipynb)`;
 - ‘report.pdf’ file : your report. Refer to the report template for the format.